

Mariachiara Cavagna

Computer Science MSc · Inditex Tech Student Ambassador · Encode Hub Scholar
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SUMMARY

Computer Science MSc student at Queen Mary University of London. MSc research extends a multimodal AI pipeline in PyTorch, working with CLAP and MuQ-MuLan audio-text foundation models. Strong focus on experimental evaluation, including a full human-subjects validation study with rigorous statistical testing.

RESEARCH & PROJECTS

MSc Research Project: Contextual and Sequential Tagging of Music Using Multimodal Embeddings

Queen Mary University of London · 2025 – 2026

- Worked with and extended an existing multimodal AI pipeline in Python (PyTorch, Hugging Face Transformers) over three months. Ran the CLAP and MuQ-MuLan audio-text foundation models across 6 model variants.
- Built a new evaluation dataset from scratch, EQSurvey, using audio segments suggested by the supervisor and clips from the SocialEQ dataset. Used it alongside the established CAL500exp benchmark to evaluate the models.
- Developed a novel context-aware tagging method (a “similarity-shift” approach). Tested it across different weightings to see what improved performance.
- Designed and ran a within-subject human evaluation study to validate model outputs against human perception. Built the experiment in jsPsych, deployed it on GitHub Pages, and recruited the participants directly.
- Ran the statistical analysis in Python using mixed effects models. Used k-means clustering to choose a varied, representative set of clips for the study.
- Worked largely independently, with guidance from the supervisor, on the tagging method, the evaluation study, and the statistical analysis.

Behavioural risk modelling

Python

- Built supervised learning models (Logistic Regression, Random Forest) to find risk factors in a behavioural dataset of over 1,800 people. Used scikit-learn, NumPy, and Matplotlib. Evaluated the models and turned their outputs into clear insights on personality-linked risk.

Gifter — AWS x The Tech Bros hackathon (1st place)

Team of four, one-day build

- Built the backend pipeline that reads a chat log and emails AI-generated gift, date, and thoughtfulness ideas. Used AWS Bedrock for the LLM, Lambda for the pipeline, S3 for input and output storage, and SES for delivery.

TECHNICAL SKILLS

Programming: Python, SQL, HTML, CSS

Machine learning & deep learning: PyTorch, Hugging Face Transformers, scikit-learn, NumPy, Matplotlib

Multimodal & foundation models: CLAP, MuQ-MuLan audio-text embeddings, embedding-space similarity, ablation studies

Experimental methods: human-subjects evaluation (jsPsych), mixed effects models, k-means clustering

Other tools: Claude Code, Vercel v0, Lovable, Rhino, Blender, Adobe Suite

EDUCATION

Queen Mary University of London

Sep 2025 – Present

MSc Computer Science · London, UK

Dissertation: Contextual and Sequential Tagging of Music Using Multimodal Embeddings.

Relevant coursework: data analytics, risk & decision modelling, database systems, systems analysis, software engineering life cycles, security protocols.

Societies: AI Society, WoSTEM, QM competitive tennis team.

Politecnico di Torino

Oct 2018 – Dec 2022

BSc Design and Visual Communication (Product Design focus) · Turin, Italy

Final thesis: designed a collaborative multitasking robot, with CIM4.0.

Earlier: Exchange year, Eric Hamber Secondary School, Vancouver (2015–2016) · Scientific Diploma, Liceo Scientifico Gobetti-Segrè, Turin (2012–2017).

PROFESSIONAL EXPERIENCE

Brandy Melville

Nov 2023 – Aug 2025

Sales and Visual Communication Manager · Turin, Italy

- Planned and published social media content. Kept the store layout appealing and offered styling suggestions. Supported brand communication and a smooth customer experience.

CIM4.0 Competence Center

Sep 2021 – Oct 2023

Product and UI Designer · Turin, Italy

- Supported an experimental AI solution presented at an international competition. Worked in multidisciplinary teams on the design, interface, and interaction logic of an industrial collaborative robot. Learned UX methods independently to deliver interactive prototypes under tight deadlines.

LANGUAGES & CERTIFICATIONS

Languages: Italian (native), English (full professional, IELTS Academic 8.0), Spanish (limited working), French (basic).

Certification: Inside LVMH Certificate, specialisation in Creation & Branding and Retail & Customer Experience. Issued November 2023.